TR2 Worked Examples: Coterminal Angles and Arc Length

Recording companion for the TR2 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

How many full turns, and how much of a turn? Coterminal angles end in the same place, so they differ by full turns: add or subtract 360° or 2π. For an arc, $s = r\theta$: the radius times the angle in radians. If the angle is in degrees, convert it to radians first; the arc is then that fraction of the way around a circle of radius r.

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Example 1: In Degrees

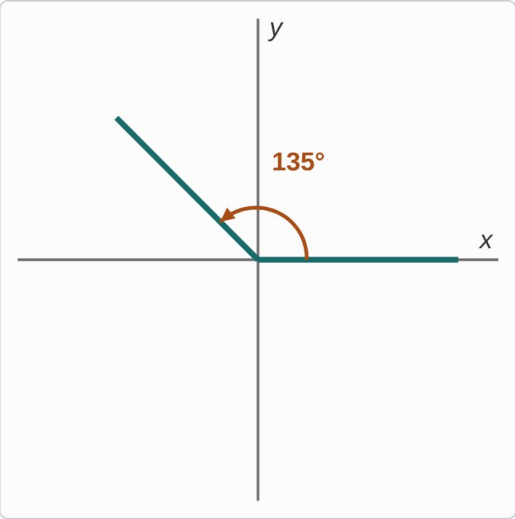


Figure: An angle in standard position: an arc labeled 135° turns counterclockwise from the positive x-axis to a terminal side pointing up and to the left, halfway between the positive y-axis and the negative x-axis.

$$135°$$

Question

Which pair is one positive and one negative angle, both coterminal with this angle?

- **A.** 495° and −225°
- **B.** 495° and −135°
- **C.** 315° and −225°
- **D.** 315° and −135°

Answer and reasoning

Add a full turn: $135°+360°=495°$. Subtract a full turn: $135°-360°=-225°$. Both end on the same terminal side as 135°. 315° adds only 180°, a half turn, which points the opposite way; $-135°$ negates the angle, which is not $135°-360°$; so 495° with $-135°$ and 315° with $-225°$ each have one slip, and 315° with $-135°$ has both. **495° and −225°.**

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Example 2: In Radians

$$4\pi/5$$

Question

Which pair is one positive and one negative angle, both coterminal with this angle?

- **A.** 9π/5 and −6π/5
- **B.** 9π/5 and −4π/5
- **C.** 14π/5 and −6π/5
- **D.** 14π/5 and −4π/5

Answer and reasoning

Write 2π in fifths: $2\pi = 10\pi/5$. Then $4\pi/5 + 10\pi/5 = 14\pi/5$ and $4\pi/5 - 10\pi/5 = -6\pi/5$. $9\pi/5$ adds $\pi = 5\pi/5$, only a half turn; $-4\pi/5$ negates the angle instead of subtracting a turn; the pairs that mix one of each have one slip, and $9\pi/5$ with $-4\pi/5$ has both. ***14π/5* and *−6π/5*.**

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Example 3: Not Coterminal

$$40°$$

Question

Which of these angles is NOT coterminal with this angle?

A. 400°
B. −320°
C. 760°
D. −40°

Answer and reasoning

Each coterminal angle differs from 40° by a whole number of full turns: $400=40+360$, $-320=40-360$, and $760=40+2(360)$, two full turns. But $-40°$ differs from 40° by 80°, not a full turn: it is the mirror image of 40° across the x-axis, in Quadrant IV instead of Quadrant I. CheckIt uses angles like 40°, $1/9$ of a full turn, that are not special angles; the full-turn rule works the same. **−40°.**

Example 4: A Negative One

$$3\pi/2$$

Question

Which is a negative angle coterminal with this angle?

- **A.** $-\pi/2$
- **B.** $-3\pi/2$
- **C.** $-7\pi/2$
- **D.** -2π

Answer and reasoning

$2\pi = 4\pi/2$, so $3\pi/2 - 4\pi/2 = -\pi/2$. That makes sense: $3\pi/2$ points straight down, and so does a quarter turn clockwise. $-3\pi/2$ negates the angle, which points straight up; $-7\pi/2$ negates the positive coterminal angle $7\pi/2$ instead of subtracting, and it points straight up too; and -2π is the full turn by itself, which points along the positive x-axis. **$-\pi/2$.**

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Example 5: Radius 6

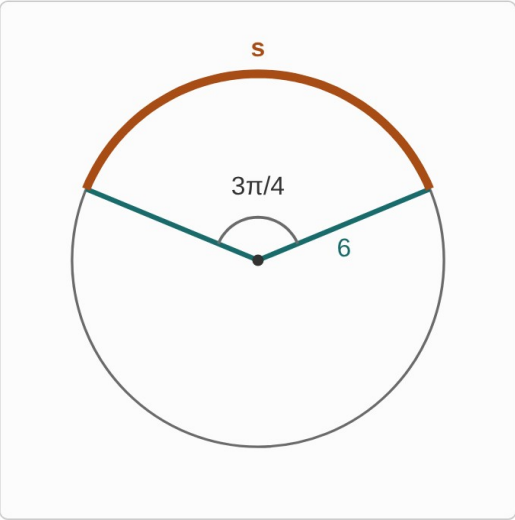


Figure: A circle with two radii, one labeled 6, forming a central angle labeled 3π/4 at the center. The arc between the ends of the radii, across the top of the circle, is highlighted and labeled s.

$$r=6, \quad \theta=3\pi/4$$

Question

What is the length of the arc cut off by this angle?

- **A.** 9π
- **B.** $9\pi/2$
- **C.** 810
- **D.** 1620

Answer and reasoning

The angle is already in radians, so $s=r\theta=6\cdot 3\pi/4=18\pi/4=9\pi/2$, about 14.1 units. A check: $3\pi/4$ is $3/8$ of a full turn, and $3/8$ of the circumference 12π is $9\pi/2$. 9π uses the diameter, 12, instead of the radius; 810 multiplies 6 by 135°, the angle in degrees; and 1620 makes both slips. ***9π/2*.**

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Example 6: An Angle in Degrees

$$r=9, \quad \theta=80°$$

Question

What is the length of the arc cut off by this angle?

A. 4π
B. 720
C. 1440
D. 8π

Answer and reasoning

Convert first: $80\cdot\pi/180=4\pi/9$. Then $s=9\cdot4\pi/9=4\pi$. 720 is $9\cdot80$, with the angle still in degrees; 8π uses the diameter, 18; and 1440 is $18\cdot80$, both slips. ***4π*.**

Example 7: Radius 14

$$r=14, \quad \theta=144°$$

Question

What is the length of the arc cut off by this angle?

- **A.** *112π/5*
- **B.** *56/5*
- **C.** *56π/5*
- **D.** *112/5*

Answer and reasoning

144·π/180=4π/5, dividing by 36. Then *s=14·4π/5=56π/5*, about 35.2. *112π/5* uses the diameter, 28; *56/5* drops the *π*; and *112/5* makes both slips. A fraction with *π* like this is normal for CheckIt, which prints it as *(56/5)π*, the same number. ***56π/5*.**

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Example 8: Part of the Circle

$$r=5, \quad \theta=3\pi/2$$

Question

What is the length of the arc cut off by this angle?

- **A.** $15\pi/4$
- **B.** $15\pi/2$
- **C.** $75\pi/4$
- **D.** $3\pi/10$

Answer and reasoning

$s=5\cdot 3\pi/2=15\pi/2$. As a fraction of the circle: $3\pi/2$ is $3/4$ of a full turn, and $3/4$ of the circumference 10π is $15\pi/2$. $15\pi/4$ takes $3/4$ of $\pi\, r=5\pi$, half the circumference; $75\pi/4$ takes $3/4$ of $\pi\, r^2$, the area, not the length; and $3\pi/10$ divides by the radius. ***15π/2*.**

Example 9: 160 Degrees

$$160°$$

Question

What is this angle in radians?

A. $4\pi/9$
B. $9\pi/8$
C. $8\pi/9$
D. $9\pi/4$

Answer and reasoning

$160\cdot\pi/180=160\pi/180=8\pi/9$, dividing by 20. 160° is a little less than 180°, and $8\pi/9$ is a little less than π. $4\pi/9$ divides by 360 instead of 180; $9\pi/8$ flips the fraction; and $9\pi/4$ makes both slips. ***8π/9*.**

Example 10: Two Fifths of Pi

$$2\pi/5$$

Question

What is this angle in degrees?

A. 144°
B. 72°
C. about 0.02°
D. 450°

Answer and reasoning

$2\pi/5 \cdot 180/\pi = 360/5 = 72°$, a fifth of a full turn. 144° uses $\pi = 360°$; about 0.02° multiplies by $\pi/180$ instead of $180/\pi$, giving $(2\pi^2)/900$; and 450° flips the fraction to $5/2 \cdot 180$. **72°.**

Example 11: 288 Degrees

$$288°$$

Question

What is this angle in radians?

A. $8\pi/5$
B. $5\pi/8$
C. $8/5$
D. $5/8$

Answer and reasoning

$288 \cdot \pi/180 = 288\pi/180 = 8\pi/5$, dividing by 36. 288° is $4/5$ of a full turn, and $4/5 \cdot 2\pi = 8\pi/5$. $5\pi/8$ flips the fraction; $8/5$ drops the π; and $5/8$ makes both slips. ***8π/5*.**

Example 12: Ten Ninths of Pi

$$10\pi/9$$

Question

What is this angle in degrees?

A. 400°
B. 162°
C. 200°
D. about 0.06°

Answer and reasoning

$10\pi/9 \cdot 180/\pi = 10 \cdot 20 = 200°$, since each $\pi/9$ is 20°. 400° uses $\pi = 360°$; 162° flips the fraction to $9/10 \cdot 180$; and about 0.06° multiplies by $\pi/180$ instead of $180/\pi$. **200°.**

Example 13: Slip 1

14π/9: 32π/9 and −14π/9

Question

A student found a positive and a negative angle coterminal with 14π/9. What is the mistake, if any?

A. There is no mistake.
B. Subtract 2π for the negative one: −4π/9.
C. The positive one should be 23π/9, adding π.
D. Both should be negative: −14π/9 and −32π/9.

Answer and reasoning

2π=18π/9, and *14π/9+18π/9=32π/9* is right. But *−14π/9* just negates the angle, so it ends in Quadrant I instead of Quadrant IV. Subtract a full turn: *14π/9−18π/9=−4π/9*. Adding *π* is a half turn, which gives *23π/9*, on the opposite side; and CheckIt asks for one positive and one negative angle, and *−32π/9* is not coterminal either. ***32π/9* and *−4π/9*.**

Example 14: Slip 2

$$s=r\theta=4(135)=540$$

Question

A student found the arc length for a radius of 4 and an angle of 135°. What is the mistake, if any?

A. There is no mistake.
B. Use the diameter instead: s = 8(135) = 1,080.
C. Convert to radians: s = 4(3π/4) = 3π.
D. Divide instead: s = 135/4 = 33.75.

Answer and reasoning

$s=r\theta$ needs θ in radians. $135°=3\pi/4$, so $s=4\cdot 3\pi/4=3\pi$, about 9.4. The whole circle is only $8\pi\approx 25$ around, so 540 is far too long. The diameter makes it worse, and the formula uses the radius; and dividing has no reason behind it. ***3π*.**

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Example 15: Slip 3

$$320°: \quad 320+360=680°, \quad 320-360=-40°$$

Question

A student found a positive and a negative angle coterminal with 320°. What is the mistake, if any?

A. There is no mistake.
B. The negative one should be −320°.
C. The positive one should be 500°, adding 180°.
D. A coterminal angle must be between 0° and 360°.

Answer and reasoning

Adding and subtracting 360° keeps the terminal side in the same place: 680° and $-40°$ both end 40° below the positive x-axis, like 320°. The student is right. $-320°$ is the mirror image, in Quadrant I; adding 180° is a half turn; and coterminal angles can be any size, which is why CheckIt asks for a negative one. **There is no mistake.**

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Example 16: Slip 4

$$s=14\cdot 4\pi/9=56\pi/9$$

Question

A student found the arc length for a circle of radius 7 and an angle of 4π/9. What is the mistake, if any?

A. There is no mistake.
B. Convert 4π/9 to degrees first: s = 14(80) = 1,120.
C. The π should cancel, so s = 56/9.
D. Use the radius, not the diameter: s = 28π/9.

Answer and reasoning

The formula is $s=r\theta$, with the radius 7: $s=7\cdot 4\pi/9=28\pi/9$. The student used 14, the diameter, which doubles the arc. A check: $4\pi/9$ is $2/9$ of a full turn, and $2/9$ of the circumference 14π is $28\pi/9$. The angle must stay in radians, not degrees; and nothing cancels the π, since the radius has none. ***28π/9*.**

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Example 17: Slip 5

$$16\pi/9=320°, \quad s=3(320)=960$$

Question

A student found the arc length for a radius of 3 and an angle of 16π/9. What is the mistake, if any?

A. There is no mistake.
B. Keep radians: s = 3(16π/9) = 16π/3.
C. 320° is wrong, since 16π/9 is really 160°.
D. Use s = πr²θ for an arc: 16π².

Answer and reasoning

The conversion to 320° is right, but it was not needed: $s=r\theta$ uses radians, and $16\pi/9$ already is. So $s=3\cdot 16\pi/9=16\pi/3$, about 16.8. The whole circle is only $6\pi\approx 18.8$ around, so 960 cannot be right. Each $\pi/9$ is 20°, so $16\pi/9=320°$, not 160°; and $\pi\, r^2$ is area, so it does not belong in a length. ***16π/3*.**

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